SET - 1

- **Q1 (CPU Scheduling):** Write a C/C++ or Java program to simulate the **Shortest Job First (SJF)** CPU scheduling algorithm. Your program should calculate and display the waiting time and turnaround time for a given set of processes with their burst times.
- **Q2 (Page Replacement):** Write a program to simulate the **Least Recently Used (LRU)** page replacement algorithm. Given a reference string and the number of page frames, compute the total number of page faults.

SET - 2

- **Q1 (Memory Allocation):** Write a program to simulate and compare the **First Fit** and **Best Fit** contiguous memory allocation algorithms. Display how processes are allocated to memory blocks and identify any unallocated processes.
- **Q2 (Disk Scheduling):** Write a program to simulate the **SCAN** disk scheduling algorithm. Given the initial head position and a sequence of track requests, compute the total head movements required.

SET - 3

- **Q1 (Deadlock Management):** Write a program to simulate the **Banker's Algorithm** for **Deadlock Avoidance**. Your program must determine if the system is in a safe state after checking available resources, maximum demand, and currently allocated resources.
- **Q2 (File Allocation):** Write a program to simulate the **Indexed File Allocation** strategy. Show how the index block maps the logical blocks of a file onto disk sectors.

SET - 4

- **Q1 (CPU Scheduling):** Write a program to simulate the **Round Robin (RR)** CPU scheduling algorithm for a user-defined time quantum. Display the scheduling chart along with the average waiting time.
- **Q2 (Page Replacement):** Write a program to simulate the **Optimal Page Replacement** algorithm. Input a page reference string and find the minimum number of page faults that occur for a fixed frame size.

SET - 5

- **Q1 (CPU Scheduling):** Write a program to simulate the **First Come First Served (FCFS)** CPU scheduling algorithm. Calculate the response time, waiting time, and turnaround time for each process.
- **Q2 (Memory Allocation):** Write a program to simulate the **Worst Fit** contiguous memory allocation algorithm. Show how blocks are allocated and calculate the remaining unallocated memory fragments.

SET - 6

- **Q1 (CPU Scheduling):** Write a program to simulate the **Priority-based CPU scheduling** algorithm (non-preemptive). Handle cases where lower numbers represent higher priority.
- **Q2 (Page Replacement):** Write a program to simulate the **First In First Out (FIFO)** page replacement algorithm. Count the total number of page hits and page faults for a given frame capacity.

SET - 7

- **Q1 (Deadlock Management):** Write a program to simulate a **Deadlock Detection** algorithm. Use a wait-for graph or resource-allocation matrix to identify if the system is currently deadlocked and list the deadlocked processes.
- **Q2 (Disk Scheduling):** Write a program to simulate the **C-SCAN (Circular SCAN)** disk scheduling algorithm. Compute the total track seek distance by wrapping around back to the beginning track.

SET - 8

- **Q1 (File Allocation):** Write a program to simulate the **Sequential File Allocation** strategy. Demonstrate how files are allocated contiguous blocks and manage space availability.
- **Q2 (Disk Scheduling):** Write a program to simulate the **LOOK** disk scheduling algorithm. Compute the total track seek movements without unnecessarily traveling all the way to the absolute end of the disk.

SET - 9

- **Q1 (Deadlock Management):** Write a program to simulate the **Banker's Algorithm** for **Deadlock Avoidance**. Your program must determine if the system is in a safe state after checking available resources, maximum demand, and currently allocated resources.
- **Q2 (Page Replacement):** Write a program to simulate the **Optimal Page Replacement** algorithm. Input a page reference string and find the minimum number of page faults that occur for a fixed frame size.

SET - 10

- **Q1 (Memory Allocation):** Write a program to simulate and compare the **First Fit** and **Best Fit** contiguous memory allocation algorithms. Display how processes are allocated to memory blocks and identify any unallocated processes.
- **Q2 (Page Replacement):** Write a program to simulate the **Least Recently Used (LRU)** page replacement algorithm. Given a reference string and the number of page frames, compute the total number of page faults.